Solution

Marks given for one correct answer = 4

Marks given for 10 correct answers = 10×4

= 40

Kalaivizhi's final score = 20

Marks reduced for incorrect answers = 40 − 20

= 20

Therefore, number of questions answered incorrectly = 20 ÷ 2 = 10

Example 1.27

A shopkeeper earns a profit of ₹ 5 by selling one notebook and incurs a loss of ₹ 2 per pen while selling of his old stock. In a particular day he earns neither profit nor loss. If he sold 20 notebooks, how many pens did he sell?

Solution

Since neither profit nor loss

Profit + Loss = 0

(ie) profit = -Loss

Profit earned from 1 notebook = ₹ 5

Profit earned from 20 notebooks = 20 × ₹ 5

= ₹ 100

Loss incurred by selling pens = ₹ 100 Loss

= −100

Loss for 1 pen = ₹ 2

loss = −2

Total number of pens sold $= (-100) \div (-2)$

= 50 pens.

Activity

Students may be divided into two groups and do the activity using "In-out box". This box takes any number as input and apply the rule to the given number and gives the answer. One group will carry out the role of finding output applying the corresponding rule. One is done in each rule for your reference.

Table (I)

Rule : Add −7

in	out
-10	-17
-7	
5	
16	
4	

Table (II)

Rule : Subtract −10

in	out
-20	-10
-13	
7	
10	
15	

Table (III)

Rule :Multiply by −5

in	out
-7	+35
-12	
+15	
18	
−5	

Table (IV)

Rule :Divide by −3

in	out
-18	+6
27	
-99	
-273	
-35	

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